

Optimizing neural network architectures using Genetic Algorithms, Particle Swarm Optimization, and Simulated Annealing

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Optimization Methods for Engineers, August 2025

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Introduction - Problem Description

- **Given** an image-based dataset with n data points $d_1, d_2, \dots, d_n \in \mathbb{R}^{H \times W \times C}$, where H, W, C represent height, width and channels respectively, and $n \in \mathbb{N}$ represents the number of samples.
- **Find** $\alpha = (\alpha_1, \alpha_2, \dots, \alpha_n)^\top \in \mathcal{D}^n$ which maximizes

$$f(\alpha, \mathcal{D}) := \lambda_2 \cdot \text{Accuracy}(\alpha, \mathcal{D}) - \lambda_1 \cdot \text{NetworkSize}(\alpha)$$

Where:

- α represents the network architecture parameters
- $\mathcal{D}$ is the dataset
- λ_1, λ_2 are weighting factors balancing size vs. accuracy
- $\text{NetworkSize}(\alpha)$ counts the total number of parameters
- $\text{Accuracy}(\alpha, \mathcal{D})$ is the classification accuracy on dataset $\mathcal{D}$

Introduction - Motivation

- For L maximum layers and C component types, exhaustive search requires $\mathcal{O}(C^L)$ evaluations
- **Consequence:** Analytical methods become computationally unviable
- **Solution:** Metaheuristic algorithms (GA, PSO, SA) provide more efficient exploration [1–5]
- **Expansion:** Consider more layers such as Conv1D/Conv2D, that were not considered in the related work

Introduction - Architecture Components

Unlike traditional approaches that only focus on fully connected layers, we extend the search space to include the following layers:

- **Convolutional Layers (Conv1D/Conv2D):** Feature extraction with learnable filters
- **Fully Connected Layers (FC):** Dense layers for classification
- **Batch Normalization:** Normalization of the values for training stability
- **Max Pooling:** Spatial dimension reduction
- **Dropout:** Regularization technique to avoid overfitting
- **Activation Functions:** ReLU, Tanh, Sigmoid

Introduction - Solution Space Characteristics

- Non-linear cost function
 $f(\alpha, \mathcal{D}) := \lambda_2 \cdot \text{Accuracy}(\alpha, \mathcal{D}) - \lambda_1 \cdot \text{NetworkSize}(\alpha)$ combines discrete network size with continuous accuracy metrics
- Mixed discrete-continuous solution space consisting of architecture topology (discrete) and hyperparameters (continuous)
- Solution space can be represented by encoding each architecture α as a variable-length chromosome
- **Naive Approach:** Exhaustive search with runtime $\mathcal{O}(C^L)$ for L max layers with C component types becomes intractable for realistic network sizes

Introduction - Project Objectives

- Investigate how **Genetic Algorithms (GA)**, **Particle Swarm Optimization (PSO)**, and **Simulated Annealing (SA)** perform when applied to the problem.
- Balance the trade-off between network size and accuracy
- Evaluate the effectiveness of different layer types in automated architecture design
- Assess the performance of these algorithms by comparing them to baseline methods, such as **Random Search (RS)** and **Grid Search (GS)**.
- Assess the performance of the algorithms across different dataset complexities

Given a image-based dataset $\mathcal{D} = \{d_1, \dots, d_n\}$

Random Search (RS)

Cost function: $f(\alpha^{(i)}) = \lambda_2 \cdot \text{Accuracy}(\alpha^{(i)}, \mathcal{D}) - \lambda_1 \cdot \text{NetworkSize}(\alpha^{(i)})$

- 1 Generate N random candidate architectures $\alpha^{(1)}, \dots, \alpha^{(N)}$
- 2 Train each network and evaluate $f(\alpha^{(i)})$ for $i = 1, 2, \dots, N$
- 3 Return $(\alpha^{(best, RS)}, f(\alpha^{(best, RS)}))$ with $f(\alpha^{(best, RS)}) \geq f(\alpha^{(i)}), \forall \alpha^{(i)}$

Grid Search (GS)

Cost function: $f(\alpha^{(i)}) = \lambda_2 \cdot \text{Accuracy}(\alpha^{(i)}, \mathcal{D}) - \lambda_1 \cdot \text{NetworkSize}(\alpha^{(i)})$

- 1 Define systematic grid over architecture hyperparameters
- 2 Evaluate $f(\alpha^{(i)})$ for each grid point $\alpha^{(i)}$
- 3 Return $(\alpha^{(best, GS)}, f(\alpha^{(best, GS)}))$ with maximal fitness

Algorithms - Genetic Algorithm

Genetic Algorithm (GA) [1]:

Fitness function: $fitness = \lambda_2 \cdot Accuracy(\alpha^{(i)}, \mathcal{D}) - \lambda_1 \cdot NetworkSize(\alpha^{(i)})$

Encoding: Variable-length chromosomes representing layer sequences

- Gene type: {Conv2D, Conv1D, FC, BatchNorm, MaxPool, Dropout, Activation}
- Gene parameters: kernel_size, stride, filters, units, etc.

Algorithm Steps:

- 1 Create population_size random architectures $\alpha^{(i)} = (layer_1, layer_2, \dots, layer_k)$
- 2 Train networks and compute fitness scores
- 3 Tournament selection with tournament_size individuals
- 4 Single-point crossover with probability crossover_prob, preserving valid layer sequences
- 5 Add/remove layers with probability mutation_add_prob / mutation_del_prob and modify layer parameters with probability mutation_param_prob
- 6 Generate new population, repeat for num_generations generations
- 7 Return $(\alpha^{(best, GA)}, fitness^{(best, GA)})$

Algorithms - Particle Swarm Optimization

Particle Swarm Optimization (PSO) [3, 4]:

Position Representation: Continuous vector $x^{(i)} \in \mathbb{R}^d$ encoding the architecture parameters

Velocity Update:

$$v^{(i)}(t+1) = w \cdot v^{(i)}(t) + c_1 \cdot r_1 \cdot (pbest^{(i)} - x^{(i)}(t)) + c_2 \cdot r_2 \cdot (gbest - x^{(i)}(t))$$

Position Update: $x^{(i)}(t+1) = x^{(i)}(t) + v^{(i)}(t+1)$

Architecture Decoding: Map continuous positions to discrete architectures:

- Layer count: $\lfloor x_1 \cdot \text{max_layers} \rfloor$
- Layer types: $\text{argmax}(\text{softmax}([x_2, x_3, \dots, x_7]))$
- Layer parameters: sigmoid/tanh transformations to valid ranges

Algorithm Steps:

- 1 Initialize swarm: Set positions $x^{(i)}(0)$ at random and set velocities $v^{(i)}(0) = 0$
- 2 Convert positions to architectures and train the networks
- 3 Update personal best $pbest^{(i)}$ and global best $gbest$
- 4 Update velocities and positions
- 5 Repeat for `num_iterations` iterations
- 6 Return $(\alpha^{(best, PSO)}, fitness^{(best, PSO)})$

Algorithms - Simulated Annealing

Simulated Annealing (SA) [2]:

State Representation: Current architecture $\alpha_{current}$

Neighbourhood Function: Small architectural modifications:

- Add/remove single layer
- Modify layer parameters
- Swap adjacent layers

Algorithm Steps:

- 1 Start with random architecture $\alpha_{current}$
- 2 Apply random modification to generate $\alpha_{neighbour}$
- 3 Acceptance Criterion:
 - If $f(\alpha_{neighbour}) < f(\alpha_{current})$: accept
 - Else: Accept with probability $e^{-\left(\frac{f(\alpha_{neighbour}) - f(\alpha_{current})}{T}\right)}$
- 4 Reduce temperature $T = T \cdot \text{cooling_rate}$ every 10 iterations
- 5 Repeat until convergence or `max_iterations`
- 6 Return $(\alpha^{(best, SA)}, fitness^{(best, SA)})$

Parametric Optimization - Variable Parameters

All algorithms run on variable parameters:

- Variable parameters for GA:

pop_size	cross_prob	mut_add	mut_del	mut_param	tourn_size	n_gen
Population size	Crossover prob.	Add mutation	Del mutation	Param mutation	Tournament size	Generations

- Variable parameters for PSO:

swarm_size	w	c ₁	c ₂	num_iter
Swarm size	Inertia weight	Cognitive coefficient	Social coefficient	Iterations

- Variable parameters for SA:

initial_temp	cooling_rate	max_iter
Initial temperature	Cooling rate	Max. iterations

- Other degrees of freedom include: the type of optimizer used in the neural network, learning rate used in training, type of crossover used in the GA, choice of r_1 and r_2 , etc.

Parametric Optimization - Fixed Parameters

- General fixed parameters used in this work:
 - $\lambda_1 = 10^{-6}$ (parameter penalty)
 - $\lambda_2 = 1.0$ (accuracy weight)
 - `max_layers` = 8
 - `epochs` = 3 (training) and 20 (evaluation)
 - `learning_rate` = 0.001
 - `optimizer` = Adam with `weight_decay` = 10^{-4}
- Algorithm-specific fixed parameters used in this work:
 - GA: `pop_size` = 40, `n_gen` = 30
 - PSO: `swarm_size` = 40, `num_iter` = 30
 - SA: `max_iter` = 1200

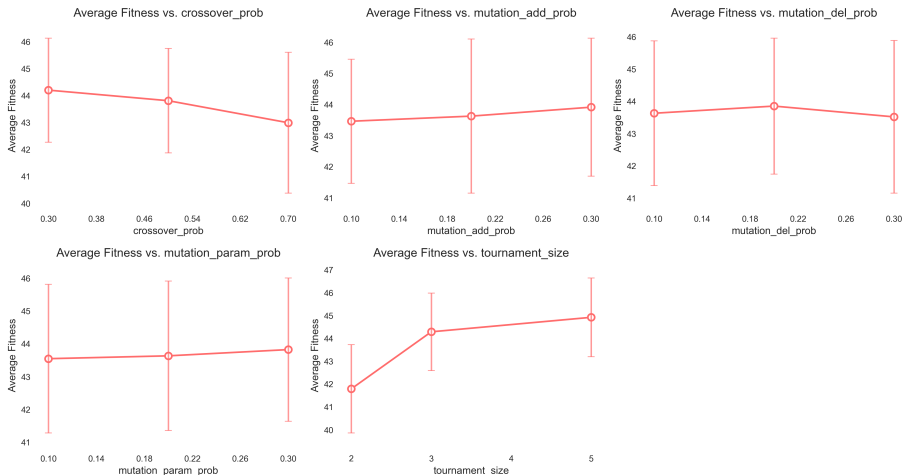
Simple algorithm to find the best parameters:

```
for each algorithm  $\in \{\text{GA, PSO, SA}\}$  do  
  for each parameter_set  $\in$  parameter_combinations do  
    for  $k = 1$  to 3 do  
      Split dataset into train/validation  
      Run algorithm with parameter_set  
      Collect: best_fitness, convergence_time, final_accuracy  
    end for  
    Compute: avg_fitness, avg_time, avg_accuracy  
  end for  
  Select parameter_set with best avg_fitness  
end for
```

Dataset (CIFAR10 / MNIST) Splits:

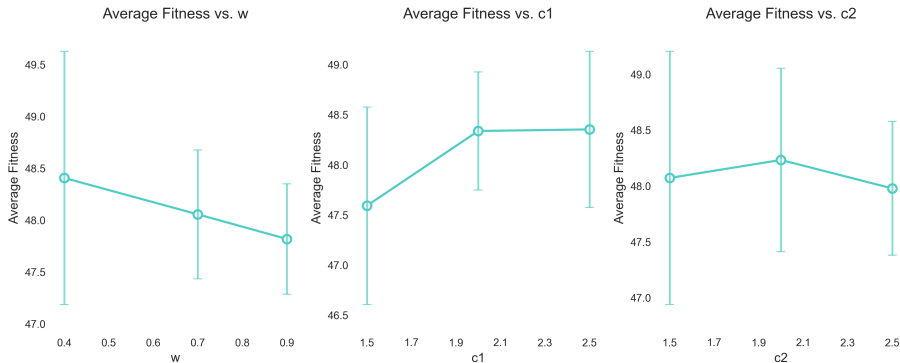
- Training: 60% (Architecture search)
- Validation: 20% (Fitness evaluation)
- Test: 20% (Final performance assessment)

Parametric Optimization - Genetic Algorithm



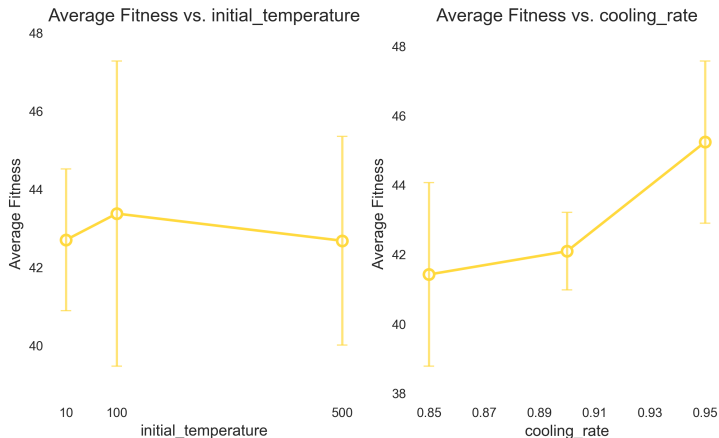
- In conclusion, the following values were chosen: $\text{cross_prob} = 0.3$, $\text{mut_add} = 0.3$, $\text{mut_del} = 0.2$, $\text{mut_param} = 0.2$, $\text{tourn_size} = 5$

Parametric Optimization - Particle Swarm Optimization



- In conclusion, the following values were chosen: $w = 0.4$, $c1 = 2.5$, $c2 = 2.0$

Parametric Optimization - Simulated Annealing



- In conclusion, the following values were chosen: `initial_temp = 100`, `cooling_rate = 0.95`

Results - Overview (MNIST)

Algorithm	Test Accuracy (%)	Parameters	Fitness	Runtime (min)
GA	97.2%	213k	95.412	7.2
PSO	97.5%	146k	97.354	14.3
SA	96.0%	27k	94.191	18.2

Table: Algorithm Performance Comparison on MNIST

- All of the tested algorithms show promising results, with a test accuracy of $\geq 96\%$
- The most accurate model was produced by PSO
- However the most accurate model is not the largest model (146k vs. 213k parameters)
- The runtime of both GA and PSO is lower than SA, as they both could be easily implemented with parallelization in mind

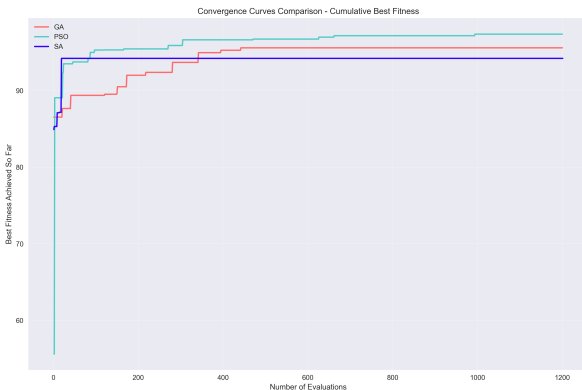
Overview (CIFAR10)

Algorithm	Test Accuracy (%)	Parameters	Fitness	Runtime (min)
GA	55.4%	703k	45.704	9.4
PSO	54.5%	56k	50.413	16.2
SA	50.3%	17k	46.858	25.2
Random	45.8%	84k	43.219	18.3

Table: Algorithm Performance Comparison on CIFAR10

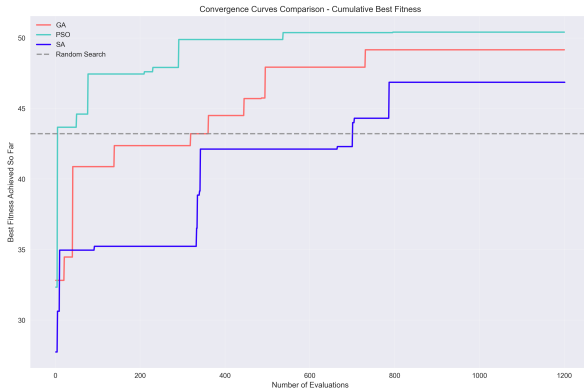
- The test accuracy is rather disappointing with the best accuracy being 55.4%
- The most accurate model is produced GA
- In this case, the most accurate model is also the largest (703k)
- Runtime higher than MNIST, which was to be expected, as CIFAR10 is a larger dataset

Results - Convergence (MNIST)



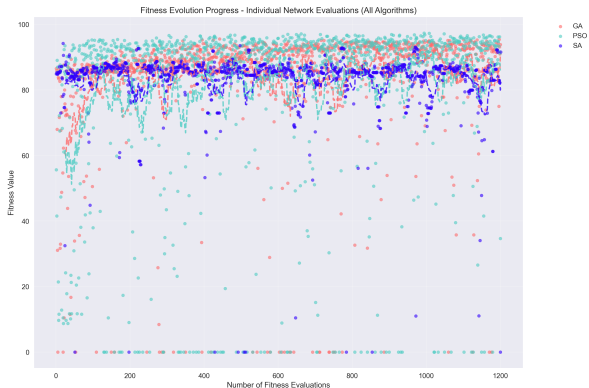
- GA shows steady convergence but plateaus early on
- PSO demonstrates steady improvement throughout iterations
- SA convergence really quickly and remains at a plateau
- Again, all algorithms converge within computational budget

Results - Convergence (CIFAR10)



- GA shows steady converges but plateaus early on
- PSO demonstrates steady improvement throughout iterations
- SA exhibits characteristic cooling behavior with gradual convergence
- Again, all algorithms converge within computational budget

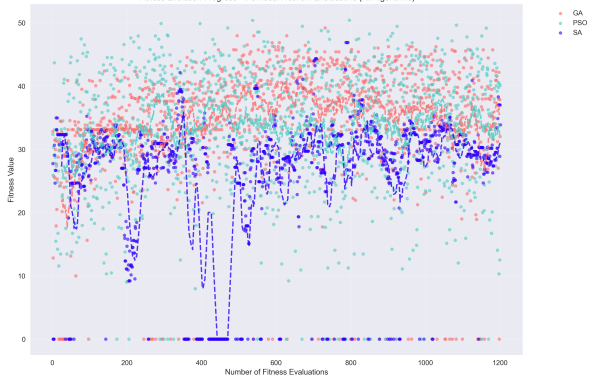
Results - Fitness Evaluation (MNIST)



- PSO converges to one swarm very quickly
- Quite a few invalid architecture edge cases at 0 fitness

Results - Fitness Evaluation (CIFAR10)

Fitness Evolution Progress - Individual Network Evaluations (All Algorithms)



- Again, PSO converges to one swarm very quickly
- Quite a few invalid architecture edge cases at 0 fitness
- SA "collapses" for a while and bounces back again

- **Genetic Algorithm (GA):**

- Crossover seems to have a negative influence
- More chance to add layers has a positive effect.
- These two things are to be expected, since larger networks should perform better

- **Particle Swarm Optimization (PSO):**

- PSO consistently performed the best for this task across most experiments and is not very volatile to parameter changes.

- **Simulated Annealing (SA):**

- SA performed the worst and can also not be parallelized like the other algorithms.
- We would not recommend using SA to optimize neural network architectures.

- Experiments were overall success and improvement over simpler methods such as Random Search and Grid Search
- CIFAR10 results were rather disappointing, but the MNIST results were good

Conclusion

- The architecture of neural networks for a specific dataset was successfully optimized using a genetic algorithm, particle swarm optimization and simulated annealing
- A hyperparameter search was conducted and we conclude that the optimization of neural networks seems quite robust to hyperparameter changes in the optimization algorithms
- Particle swarm optimization and genetic algorithms present promising pathways to optimize neural network architecture, while simulated annealing seems less promising, especially for larger experiments, as it can't be parallelized
- While these results were promising, the immense compute cost severely limits the types of applications these optimization methods could be applied to given current compute limitations
- We think the optimization of neural network architecture makes the most sense for somewhat simple problems that require both accuracy and a short inference time, such as collision avoidance for drones

Future Work and Limitations

Current Limitations:

- Limited to relatively simple architectures (≤ 10 layers)
- Training epochs are reduced for computational feasibility
- Single-objective scalarization of multi-objective problem
- Runtime of certain approaches became infeasible very quickly

Future Extensions:

- Generalize the approach to different datasets (graph, audio, video, etc)
- Introduce more layers to be considered when constructing the NNs
- Consider advanced components, such as attention, transformers, etc.
- Leverage pre-trained components in search

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